

UCF Physics: AST 5765/4762 (Advanced) Astronomical Data Analysis

Fall 2020 VirtualBox Ubuntu Installation

You can install a complete Linux OS distribution inside a virtual environment under any operating system. Linux users who are on the same version as used in the class may just install our versions of the add-on software, or may use VirtualBox and install another Linux inside it.

Ubuntu Linux 20.04.1 LTS

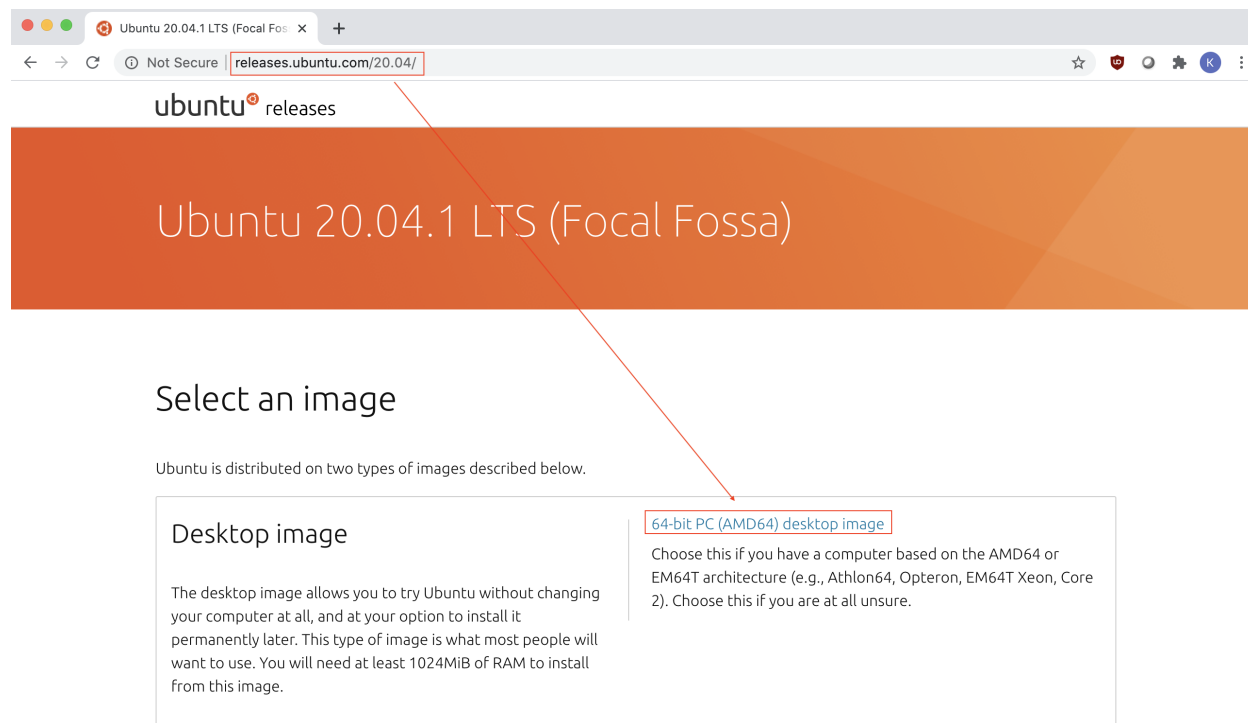
To get started, download a copy of the Ubuntu Linux installation DVD image. We will use the latest version of Ubuntu 20.04.1 LTS (Long-Term Support).

Go to this site:

<http://releases.ubuntu.com/20.04/>

Download this file:

`ubuntu-20.04.1-desktop-amd64.iso`



If that doesn't work for any reason, try the main Ubuntu website:

<https://www.ubuntu.com/download/desktop>

It may start downloading right away, depending on your browser. To skip the ad, you may need to click: [download now](#).

Alternatively, you can use an image on a DVD or USB drive, perhaps obtained from a colleague

(just be sure it is the latest version).

All machines made in the last 10 years have 64-bit architectures, but some people use a 32-bit operating system (OS) for compatibility with older software. If you think your machine is not 64-bit compatible, please check under the Control Panel/System Preferences to confirm, then talk to course staff if it really is in 32 bits.

You don't need to open this file when it is done downloading; we will use it within the Virtual-Box environment.

BIOS Setting

Mac users can skip this section.

We need to make sure your operating system has **virtualization** turned on. This lets the CPU run a guest operating system (Linux in this case) as well as the host OS.

1. Save your files and shut down your computer.
2. Boot your computer. When you see the first image (usually the manufacturer's logo) hold down the DEL key. It should bring you to the BIOS menu.
3. If it doesn't bring you to the BIOS menu, try again, holding down ESC, F12, F11, F2, or a dedicated key (e.g., Lenovo's "Think" key), in turn. If those don't work, look at your computer's instruction manual.
4. In the BIOS, locate the **virtualization** option(s). Usually these are under the first or second menu.
5. **Enable** all the options related to virtualization (check them, highlight them, whatever your BIOS uses to indicate it's turned on).
6. Save and exit.

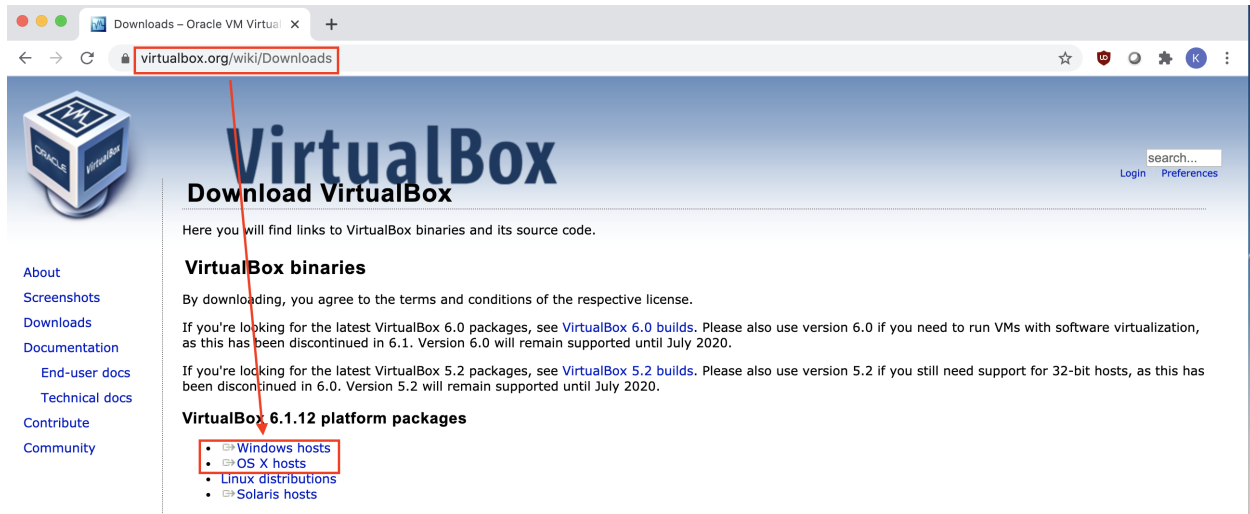
VirtualBox

VirtualBox lets you run any OS inside a program running on your native or "host" OS.

It is available at:

<https://www.virtualbox.org/wiki/Downloads>

Follow the instructions for your OS version.



If the install fails on MacOS, you may need to adjust your Security and Privacy settings. First check if apps are allowed from “App Store and identified developers”. You may need to “Allow” VirtualBox to install. If that doesn’t work, in the terminal (on your Mac) type:

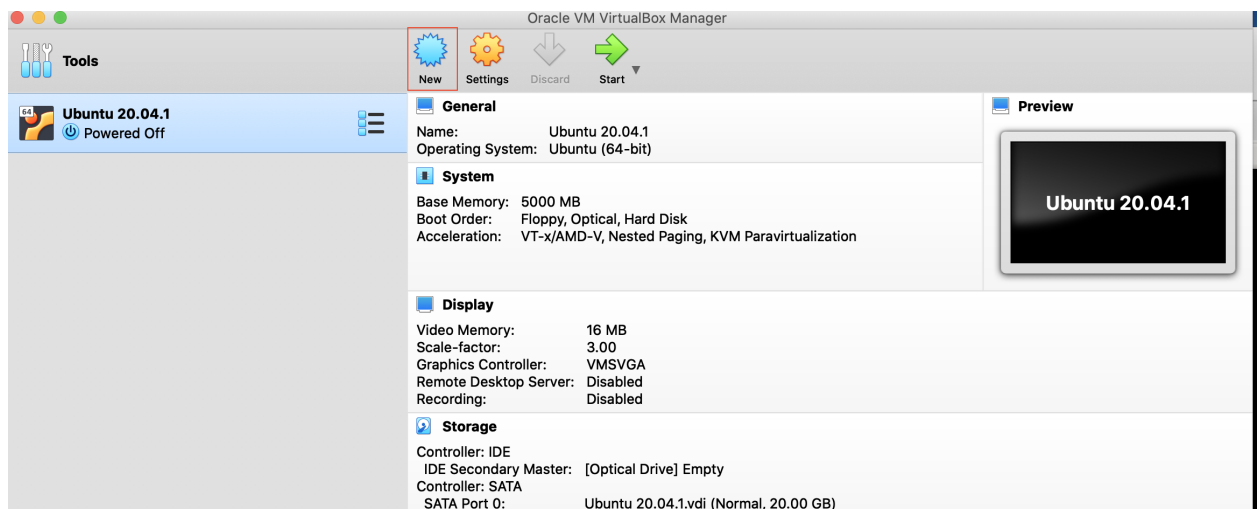
```
sudo spctl --master-disable
```

and enter your password. Then, restart System Preferences and under Security and Privacy you should now see an option to allow apps downloaded from “Anywhere”. Select this, and if necessary “Allow” VirtualBox to install. You can change your settings back after this install. If this still does not work, ask the professor or TA.

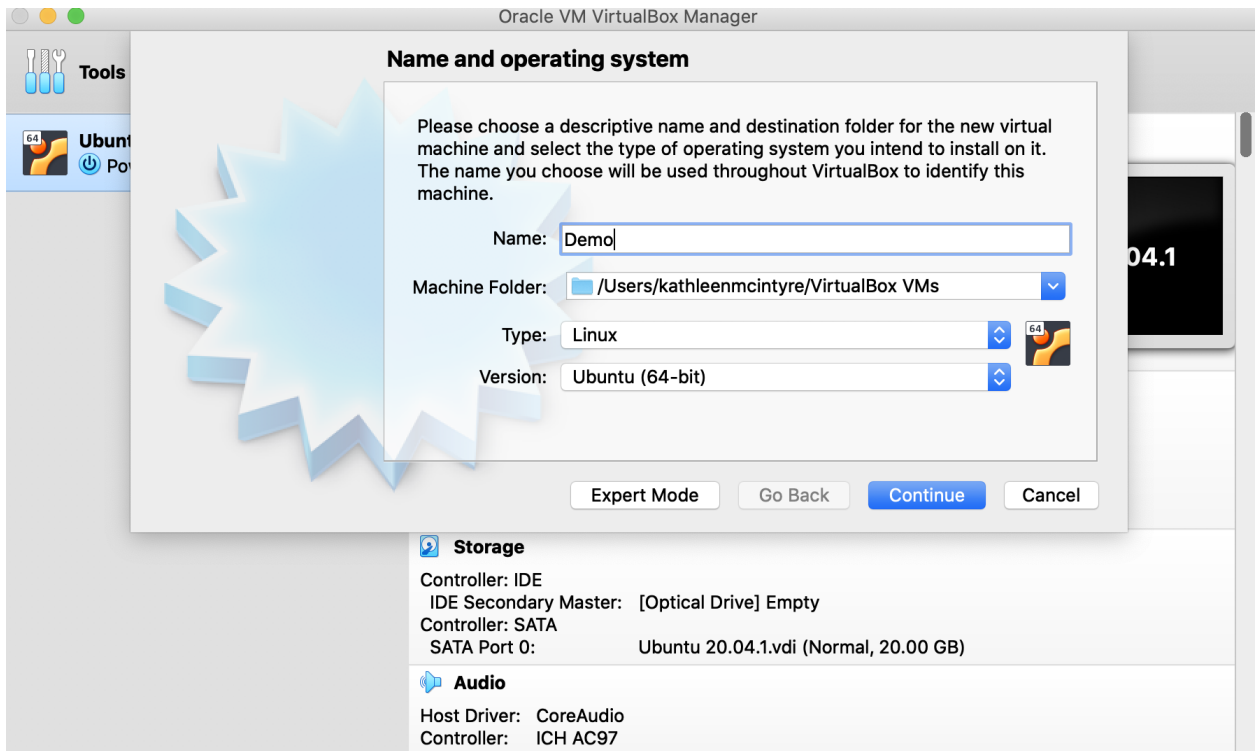
For operating systems that ask you questions during the install, leave all options at their defaults, except the installation location, if you prefer to install elsewhere.

Run VirtualBox. This will put up a dialog:

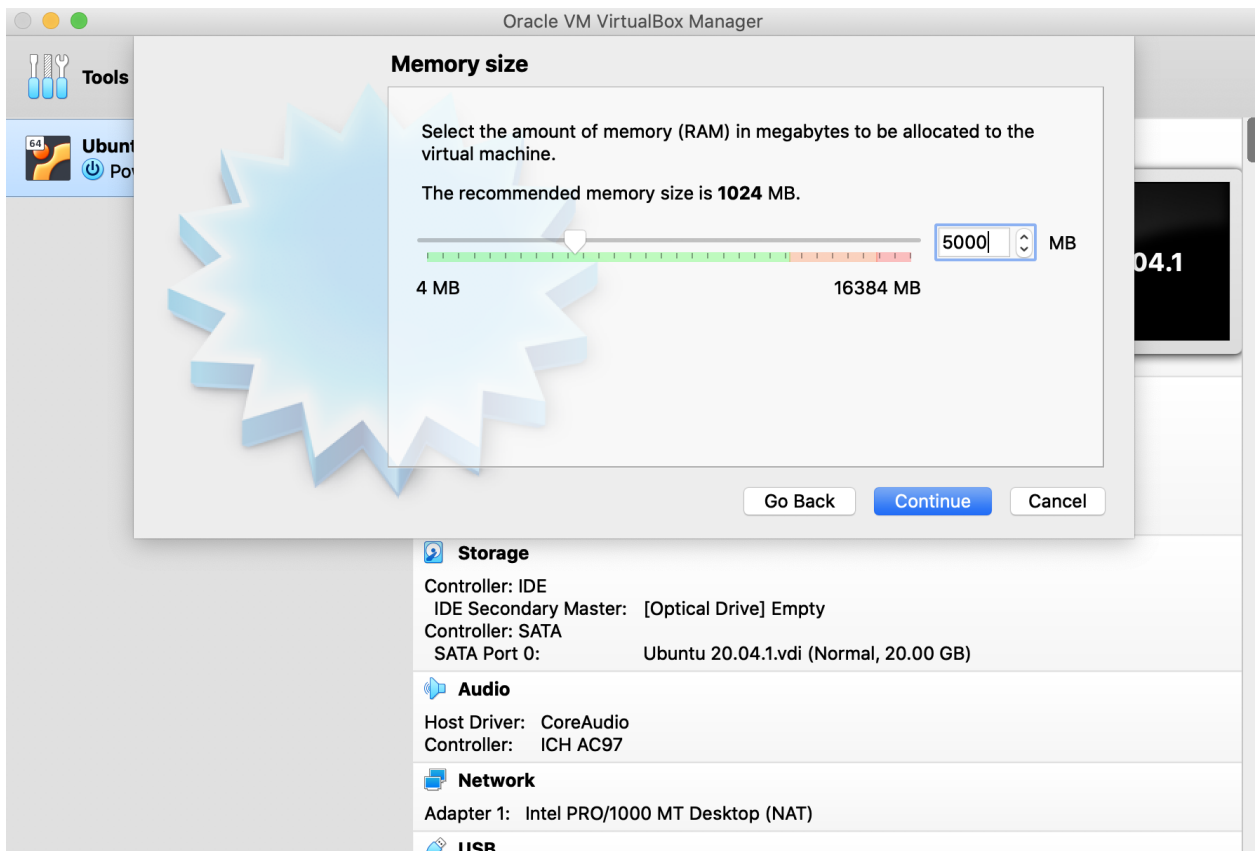
1. Click “New”.



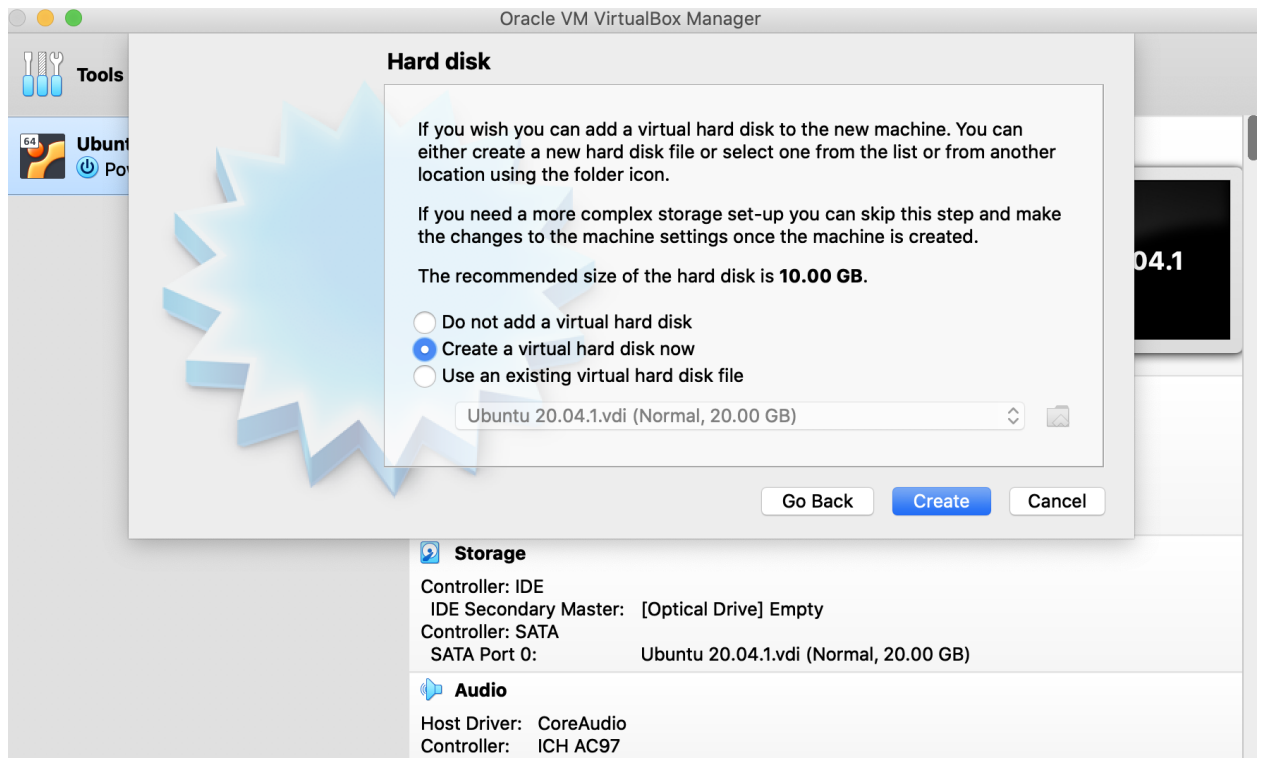
2. Choose a *short name*, such as “Linux” or “Ubuntu” or the course number.
3. Choose “Linux” as the operating system, and “Ubuntu (64-bit)” as the version. If 64-bit is not an option, go back to the instructions on BIOS settings. Failing that, talk to course staff.



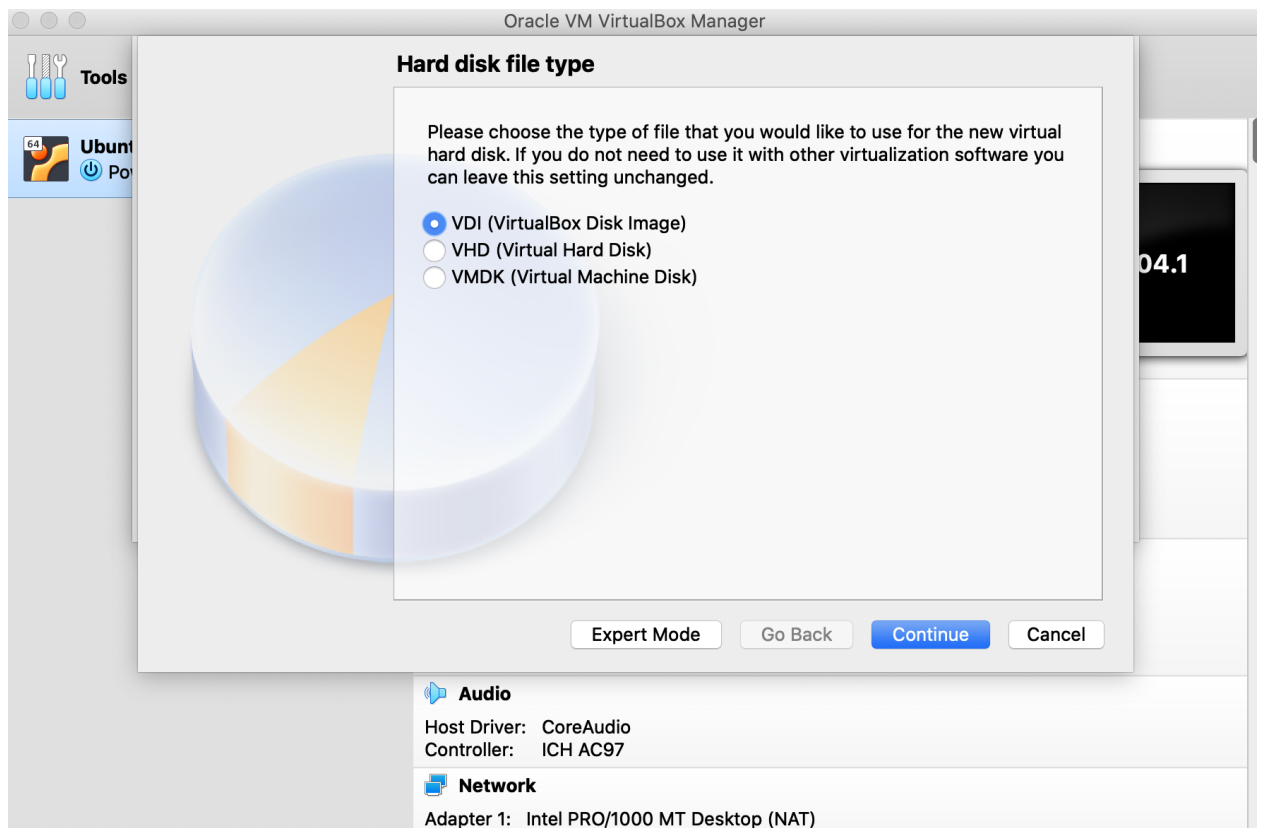
- Set the memory size to at least 3 GB, and preferably 5 GB. If 3 GB is in the red zone for your computer, talk to course staff.



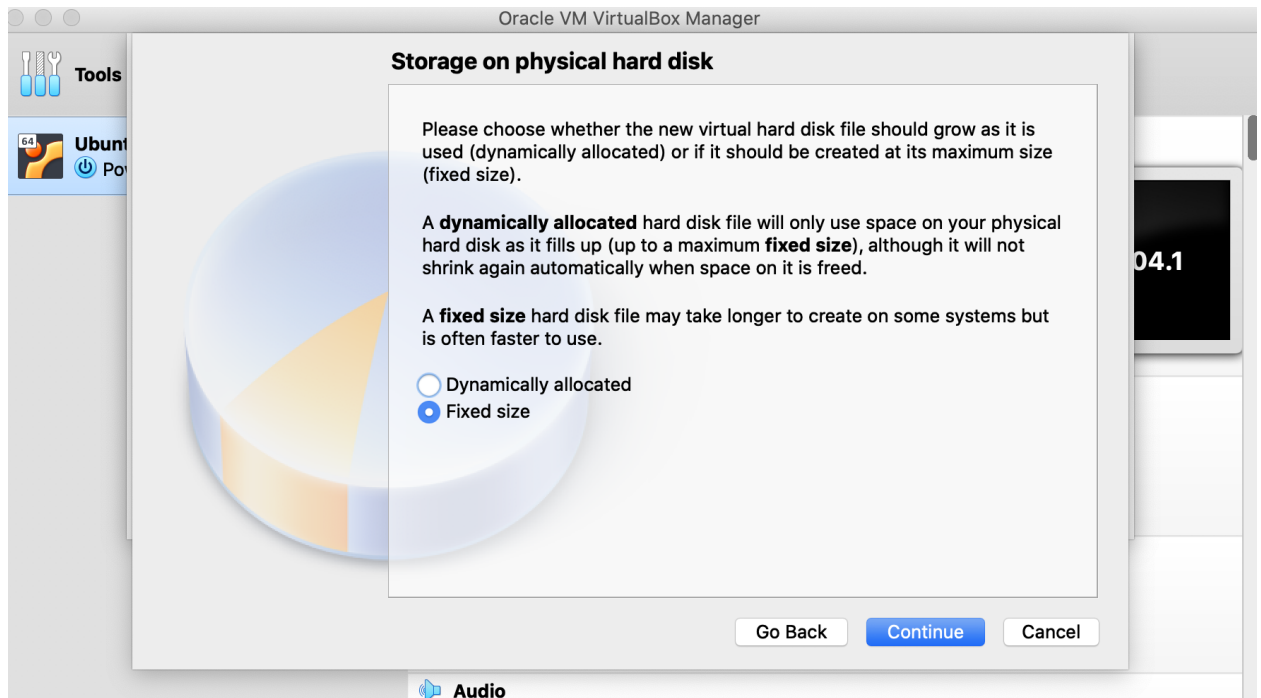
5. Choose “Create a virtual hard disk now”



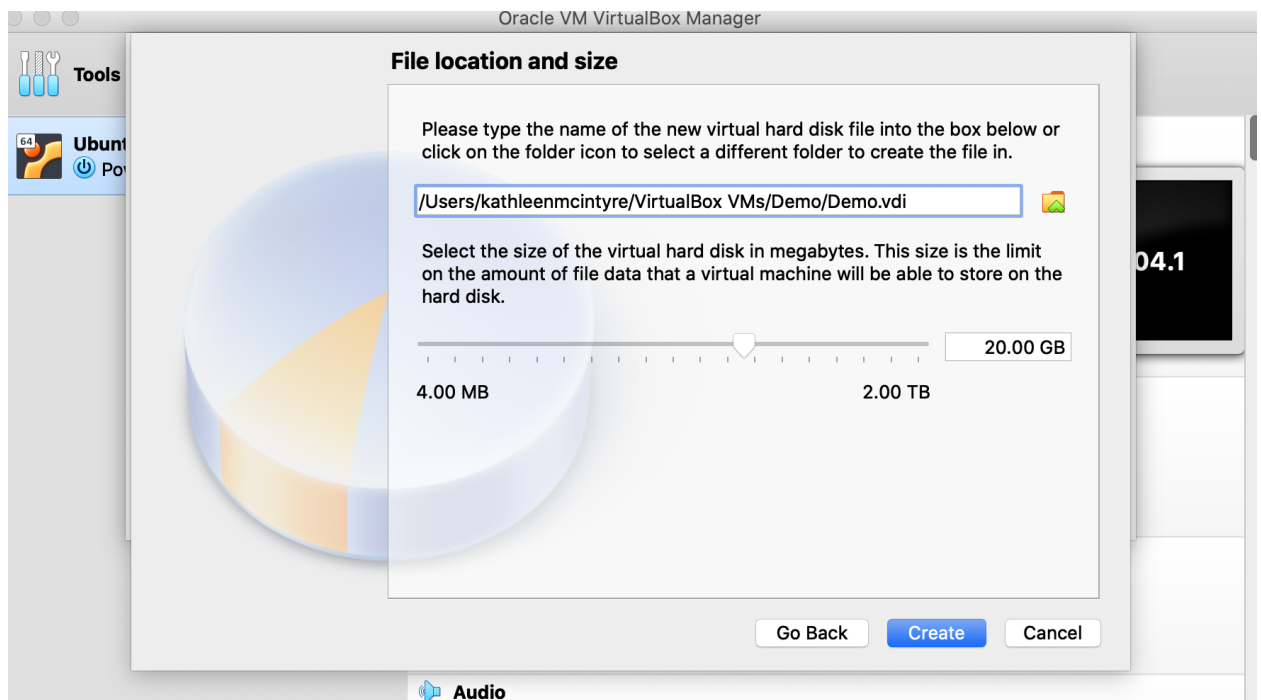
6. Choose “VDI”



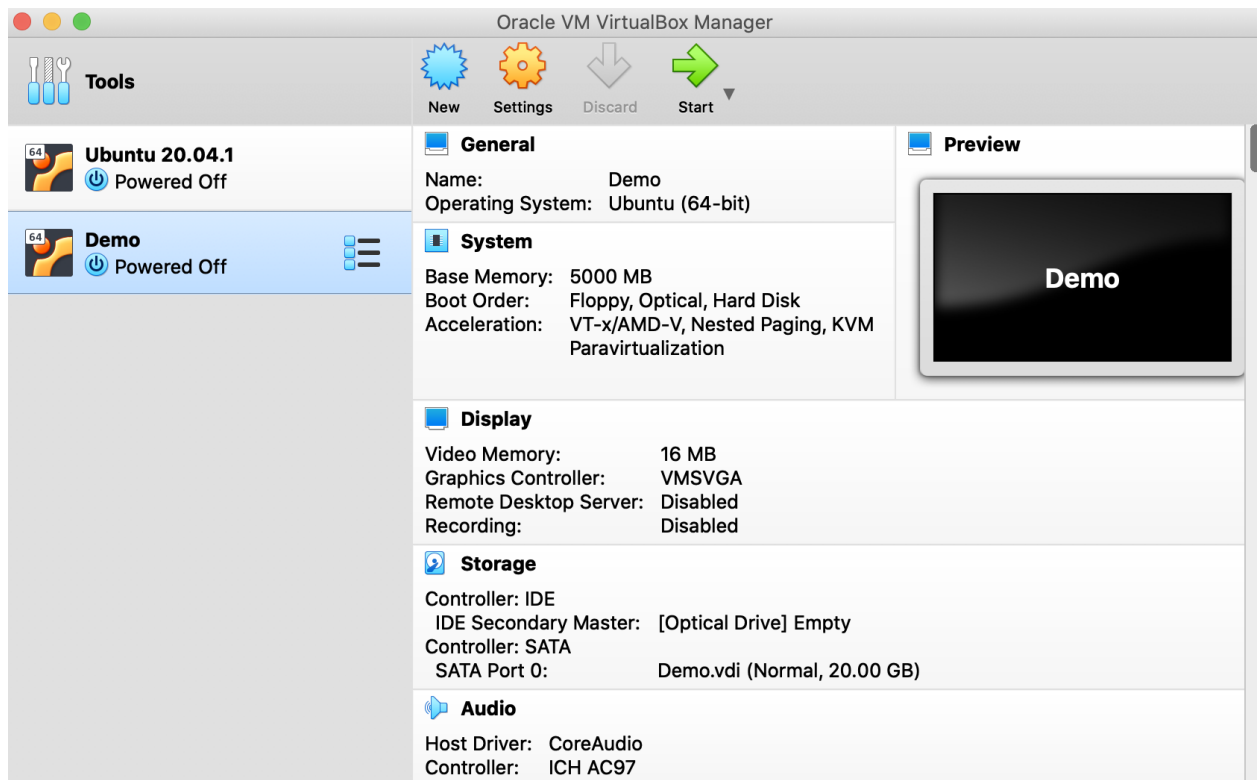
7. Choose “Fixed size” (We have seen performance issues with Dynamical, and Fixed ensures you won’t run out of space during the class.)



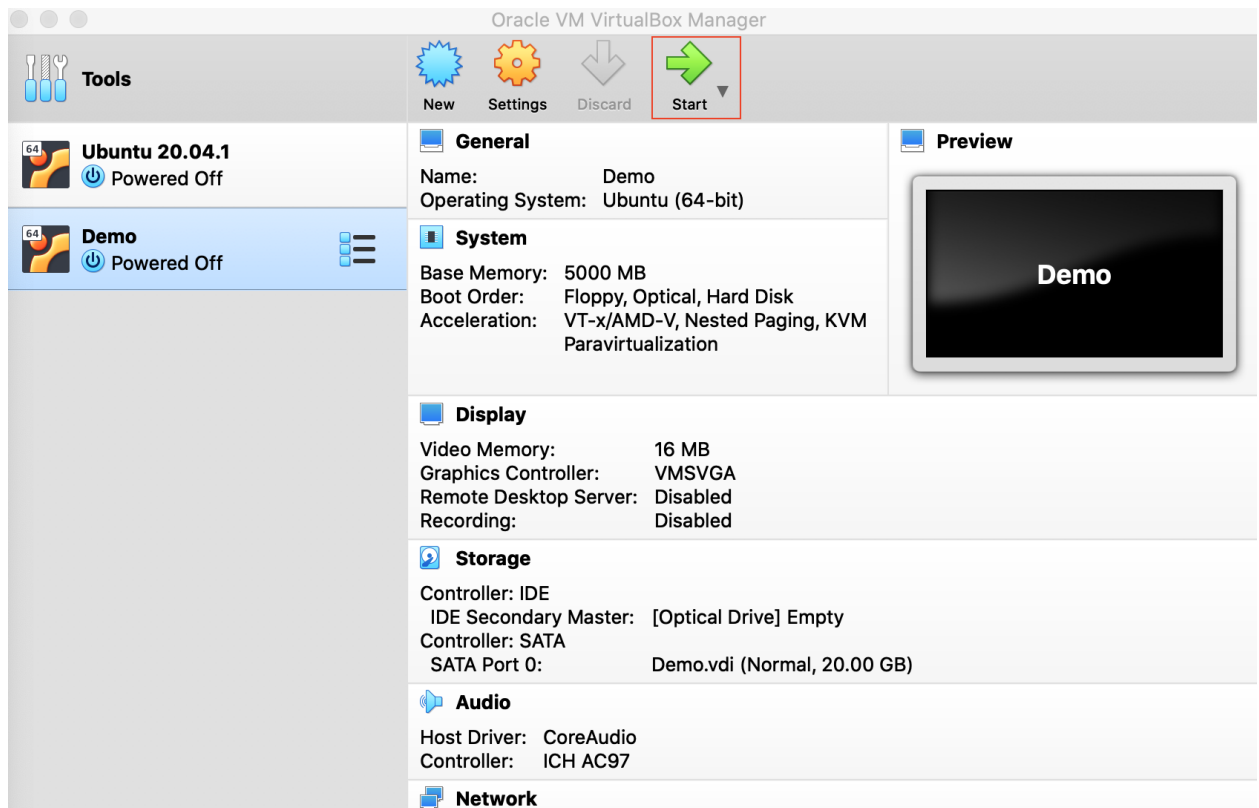
8. Name the hard disk. You could use the course number.
9. Make the size 20 GB. It will allocate that much space, which will take a few minutes.



10. In the VirtualBox Manager, single-click on your new virtual computer on the left to select it.



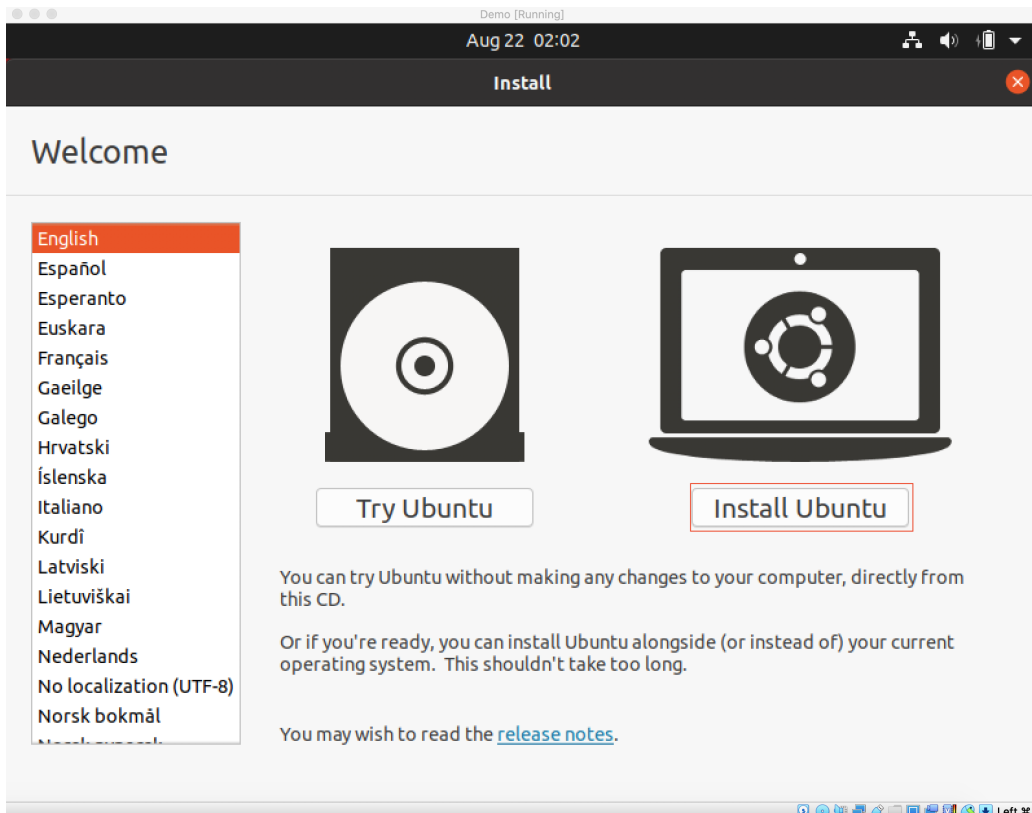
11. Click “Start”



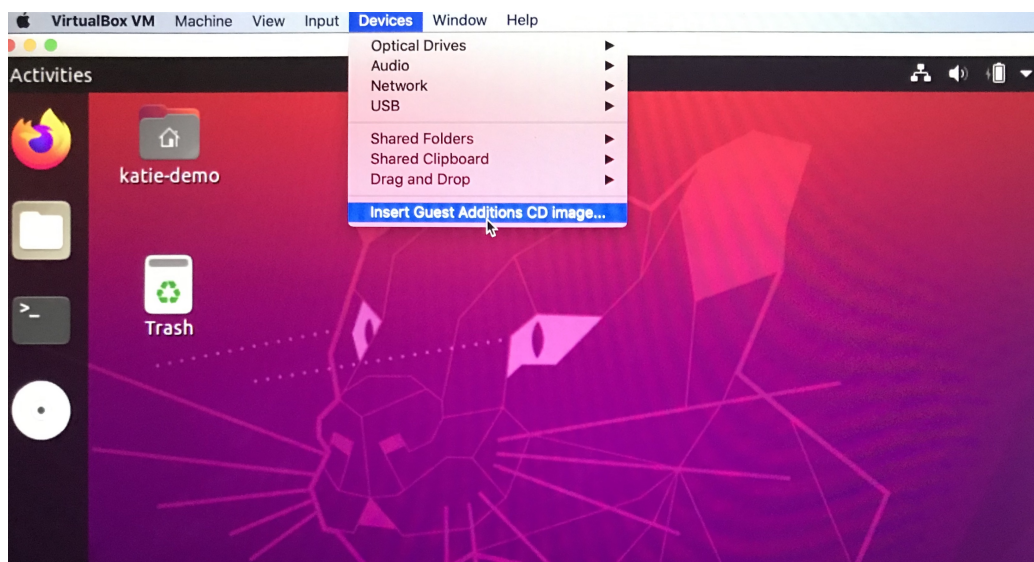
12. Click the Folder-browse icon, and navigate to the saved Ubuntu DVD file. Click “Start”. You should see “Ubuntu” and some dots on the screen.



13. In a minute or so, it should give you the option to “Install Ubuntu”. Click it.



14. Note: If you see a corrupted display (lots of colors and letters), try pressing (right ctrl - F7) and then (right ctrl - F1).
15. Choose your keyboard layout (usually the first option for US English). Click “Continue”.
16. Select “Normal installation. Check “Download updates” and “Install third-party software....”. Click “Continue”.
17. Check “Erase disk and install Ubuntu” (it “erases” the 20 GB virtual disk made above). Click “Install Now” and “Continue”.
18. Choose the New York time zone. Click “Continue”.
19. Enter your full name or a nickname for “Your Name”.
20. Choose a short computer name that isn’t your username, such as “fredvbox” or “suelinux”. This name will appear in your shell prompt, so long isn’t good. VirtualBox’s recommendation of “username-VirtualBox” is way too long!
21. Pick a short username, **based on your name**. This should be up to 7 characters long, all lowercase. The whole class will see this name in peer evaluations of your assignments. So will the TA, while grading. Please base it on your name such as joeh, jharr, jh, joe, etc. Do not include special characters or use your videogame name. Be professional.
22. Pick a strong password, check “Require my password”
23. Click “Continue” and wait for the installation to complete. Click to restart, and when prompted, hit “Enter”.
24. Under the Devices menu at the top, select “Insert Guest additions...”. It will prompt for your password. This installs the software that lets Linux know how to talk to your host OS. When it prompts, you can hit “Enter” to close the window and restart to apply the changes.

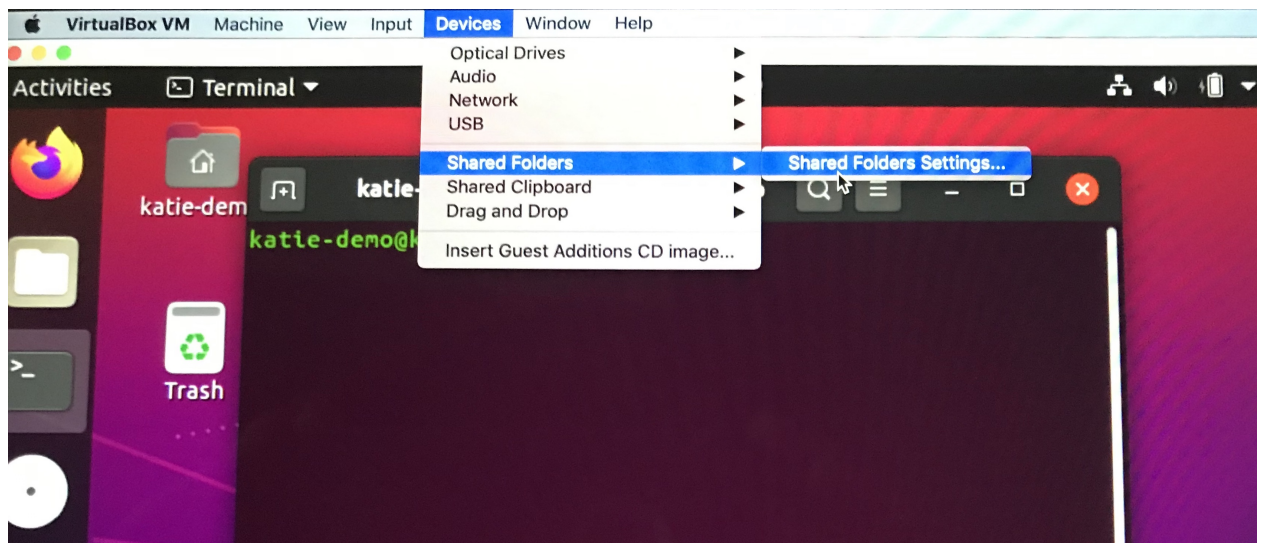


25. Start a terminal with right-click->Open Terminal.
26. Type: `sudo adduser <username> vboxsf`, replacing `<username>` with your actual username.
27. `sudo apt update` This updates the database of available packages. Always do this before installing software to get the latest versions.
28. `sudo apt upgrade` This actually does the update. It shouldn't need to update anything, because you checked the option to upgrade as it installed. However, do these two commands periodically to get security updates.
29. *Turn off the screensaver.* It will blank the screen and require a password after five minutes of inactivity. To turn this off, click the down arrow in the top-right corner. In the box that appears, click the button in the lower left. This brings up Settings. Select Power->Blank screen->Never.
30. *Send the shutdown signal.* Click the "red x" in the upper left corner. You will typically want to "Send the shutdown signal" when exiting virtual box. "Power off the machine" is the equivalent of pulling the plug. "Save the machine state" is like going into sleep mode.

VirtualBox Configuration

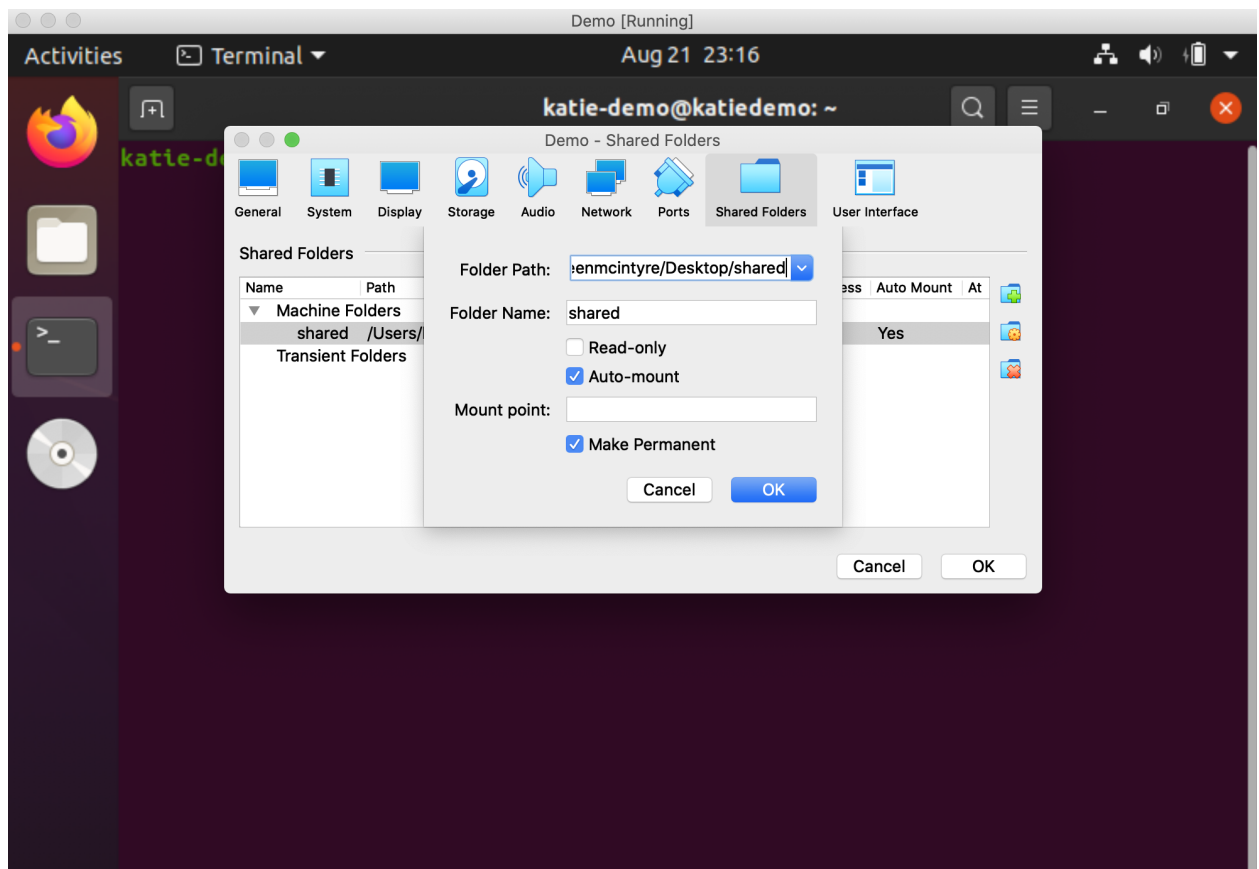
In VirtualBox, configure a shared folder, so you can move files between your native OS and the VirtualBox Linux:

1. Restart the virtual box.
2. Under the Devices menu at the top, scroll down to "Shared Folders" and select "Shared Folder Settings".



3. A dialog box will pop up. On the right, click the folder-add icon.

4. In your host OS, make a “shared” folder for files to transfer between the two OSes. It will appear in both operating systems. Think of it as a portal between worlds!
5. In the first box, enter the path to that folder.
6. The second box should autofill with the Folder Name.
7. Select Auto-mount
8. Select Make Permanent
9. Click “OK”



10. The shared folder will be located under /media/sf_shared.

Turn on mouse integration. This will let you move your mouse between VirtualBox and your desktop and have it work in both places as you expect. Use the menus at the top of the VirtualBox OS window and select:

```
Input->Mouse Integration
Devices->Shared Clipboard->bidirectional
Devices->Drag and Drop->bidirectional
```

[NOT RELIABLY WORKING YET]

Backup! It's critical. [Work in progress...tar your class directory and mail it to yourself or upload to a site like Dropbox using Firefox.]